More Experiments

1 OPAD/OPAD+ versus RB/WR

1.1 Conceptual comparison

Rao–Blackwellization (RB) and waste-recycling (WR) Monte Carlo [2, 1] are variance reduction techniques for Metropolis–Hastings that replace the contribution of each transition by its conditional expectation:

$$\alpha(x_t, x_t')f(x_t') + (1 - \alpha(x_t, x_t'))f(x_t),$$

where x_t is the current state, x_t' is the proposal, and $\alpha(x_t, x_t')$ is the acceptance probability.

This induces an effective weighting over (unique) states given by

$$w_{\mathrm{RB/WR}}(x) \propto \sum_{t:x_t=x} (1 - \alpha_t) + \sum_{t:x_t'=x} \alpha_t,$$

which depends on the proposal mechanism and acceptance probabilities. In contrast, OPAD assigns weights

$$w_{\mathrm{OPAD}}(x) \propto \tilde{p}(x), \quad x \in D,$$

where D is the set of unique visited states, and OPAD+ extends this to include additional states (e.g., proposals).

The two approaches are fundamentally different. RB/WR: (i) does not discard duplicate states, (ii) assigns weights based on transition dynamics, and (iii) does not explicitly depend on target probabilities. In contrast, OPAD/OPAD+: (i) collapses duplicate states, (ii) does not rely on the transition probabilities α_t and (iii) assigns weights using the (unnormalized) target probabilities.

RB/WR is most closely related to OPAD+, as both incorporate proposed (not necessarily accepted) states. However, OPAD+ differs in that it assigns weights independent of the sampling trajectory.

Importantly, our theory shows that over the support of accepted and proposed states D, OPAD+ yields the KL-optimal weights; therefore, OPAD+ provides a provably better approximation than any alternative weighting, including RB/WR (defined on the same set D).

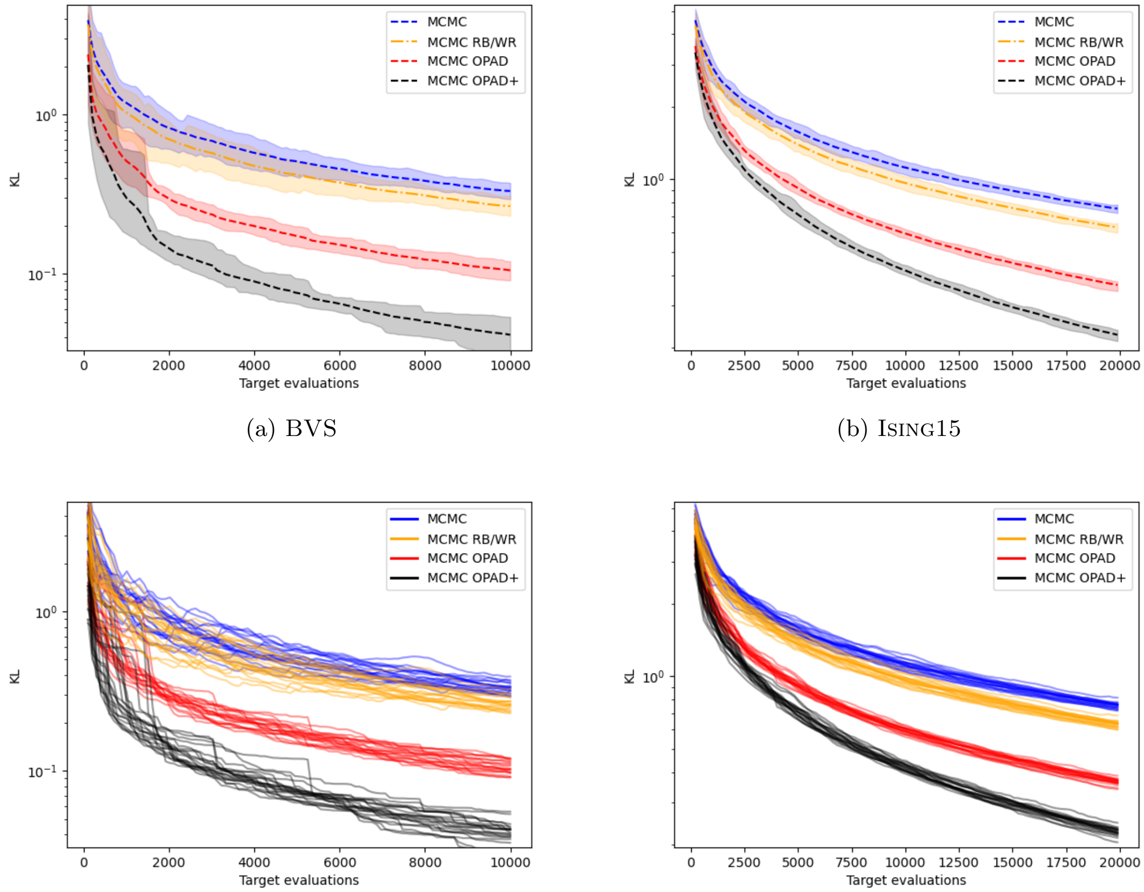
Furthermore, in terms of MSE and variance, OPAD/OPAD+ is expected to outperform RB/WR. The dominant source of variance in MCMC-based estimators arises from approximating probabilities via empirical frequencies (i.e., multiplicities), which both MCMC and RB/WR rely on. OPAD/OPAD+ removes this source by replacing frequency-based weights with model-based weights.

1.2 Experimental results

Figure 1 shows KL divergence as a function of target evaluations. The results indicate that RB/WR improves over standard MCMC, but only marginally. In contrast, OPAD and especially OPAD+ provide significantly lower KL divergence. The ordering from highest to lowest divergence is consistently:

$$\text{MCMC} \ > \ \text{RB/WR} \ \gg \ \text{OPAD} \ > \ \text{OPAD+}.$$

Figure 2 reports MSE, absolute bias, and variance. Across all Ising models, RB/WR and standard MCMC are visually indistinguishable, with curves and confidence intervals effectively overlapping. In the BVS model, RB/WR exhibits only a negligible smoothing effect over MCMC, with differences barely observable. In contrast, OPAD and OPAD+ achieve substantially lower MSE, primarily due to variance reduction.



(a) BVS

(b) Ising15

(c) BVS

(d) Ising15

Figure 1: KL divergence vs. target evaluations. Algorithms: MCMC(blue), MCMC RB/WR (orange) and MCMC OPAD/OPAD+ (red/black). Models: BVS(a & c) and Ising15(b & d) based on 20 independent MCMC chains. The mean (dashed line) and 95% confidence intervals are shown in (a) & (b), and the chains are plotted in (c) & (d).

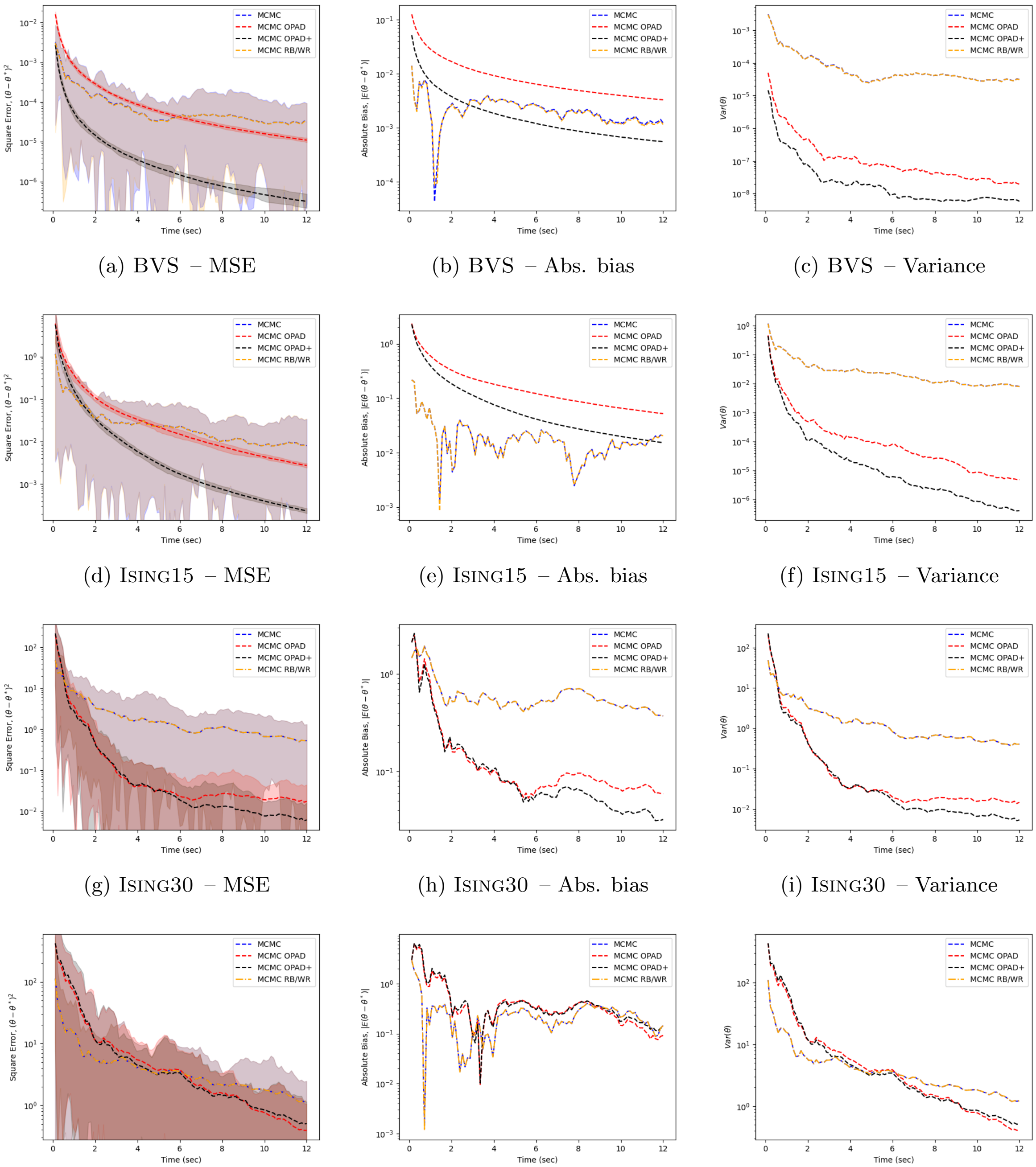
Figure 3 shows the $\widehat{R}$ diagnostic. Again, RB/WR closely tracks standard MCMC, with curves effectively overlapping across all models, indicating negligible practical improvement over MCMC.

Overall, these results demonstrate that while RB/WR provides modest variance reduction at the transition level, its practical impact is limited. In contrast, OPAD/OPAD+ yield substantially greater improvements across all metrics.

References

[1] Randal Douc and Christian P Robert. A vanilla rao–blackwellization of metropolis–hastings algorithms. 2011.

[2] D Frenkel. Waste-recycling monte carlo. In *Computer Simulations in Condensed Matter Systems: From Materials to Chemical Biology Volume 1*, pages 127–137. Springer, 2006.



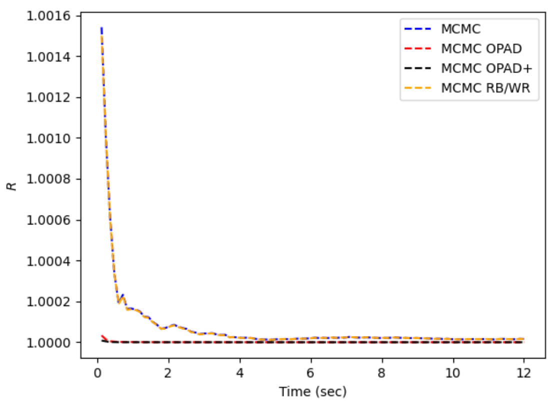
(a) BVS – MSE (b) BVS – Abs. bias (c) BVS – Variance

(d) Ising15 – MSE (e) Ising15 – Abs. bias (f) Ising15 – Variance

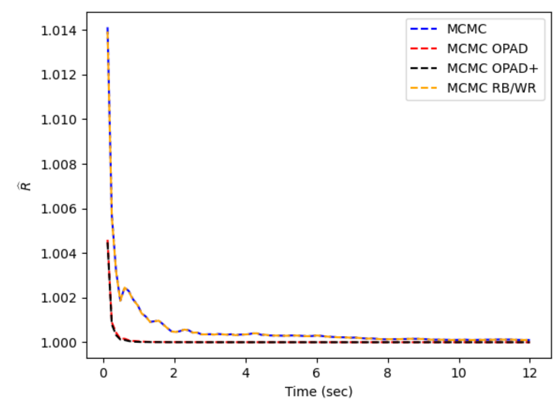
(g) Ising30 – MSE (h) Ising30 – Abs. bias (i) Ising30 – Variance

(j) Ising40 – MSE (k) Ising40 – Abs. bias (l) Ising40 – Variance

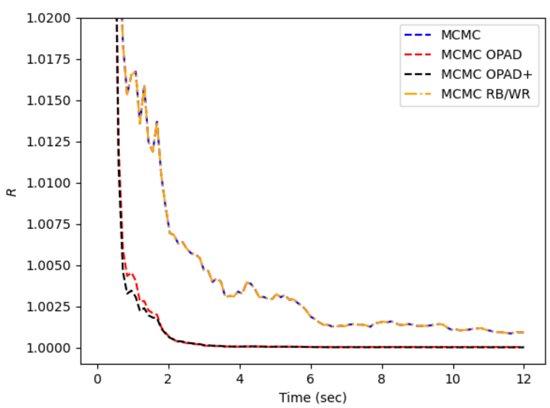
Figure 2: MSE $\pm 95\%$ confidence intervals (left column), absolute bias (middle middle column), and estimated variance (right column) for 20 independent chains as a function of MCMC chain running time. Results are shown for the baseline MCMC (blue), its OPAD/OPAD+ (red/black) and RB/WR extensions. The target models are BVS, Ising$\{15, 30, 40\}$ (rows 1-4, respectively).



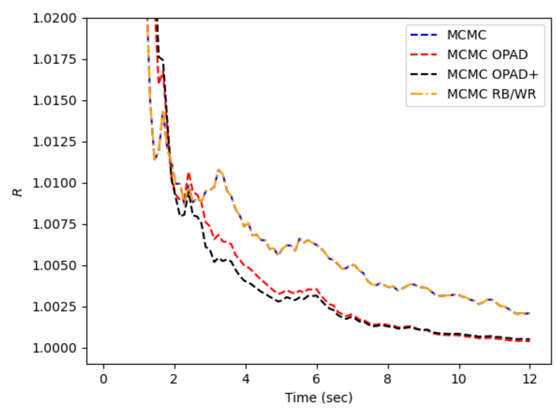
(a) $\widehat{R}$ – BVS



(b) $\widehat{R}$ – Ising15



(c) $\widehat{R}$ – Ising30



(d) $\widehat{R}$ – Ising40

Figure 3: $\widehat{R}$ versus MCMC sampling time based on 20 independent runs. Target models: BVS(a), Ising15 (b), Ising30 (c) and Ising40 (d). Algorithms: MCMC (blue), its OPAD/OPAD+ and RB/WR extensions (red/black/orange).